Zatomic

ZECATHON · CROSS-CHAIN · P2P OTC

Swap shielded ZEC for ETH.

No bridge. No wrapped ZEC.

An atomic swap for two parties who have already agreed a trade.

162-55-186-88.sslip.io · github.com/yugocabrio/zatomic

THE PROBLEM

Why a shielded atomic swap is hard

01

Neither side wants to go first

Whoever pays first can be cheated, so both payments must happen, or neither.

02

Shielded notes have no scripts

No hash locks or time locks: a shielded note moves only with a valid signature from its key.

03

The usual workarounds leak

Bridges prove Zcash history to Ethereum; other routes use solvers, public viewing keys or transparent ZEC.

> Zatomic keeps the ZEC shielded and asks Ethereum to verify **no Zcash history**.

HOW IT WORKS

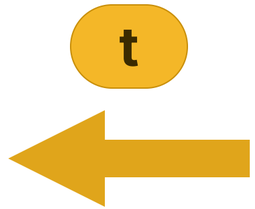
One secret settles both chains

ZCASH · IRONWOOD

Joint shielded note

2-of-2 FROST spend key

Claim C → Bob, pre-signed **ŝ**



public witness

ETHEREUM

PointEscrow

ETH → Alice

checks **t·G = T** on Pallas

> Alice claims the ETH with t. The same t completes Bob’s Zcash signature: **s = ŝ + t**.

PREPARED BEFORE FUNDING

Both exits exist before funding

IF ALICE REVEALS T

Both get paid

Alice withdraws the ETH. Bob reads t from Ethereum and completes his shielded ZEC claim, even with Alice offline.

IF NOBODY REVEALS

Both get their own asset back

Bob refunds his ETH after the deadline. Later, Zcash’s own lock time lets Alice’s pre-signed refund return her ZEC.

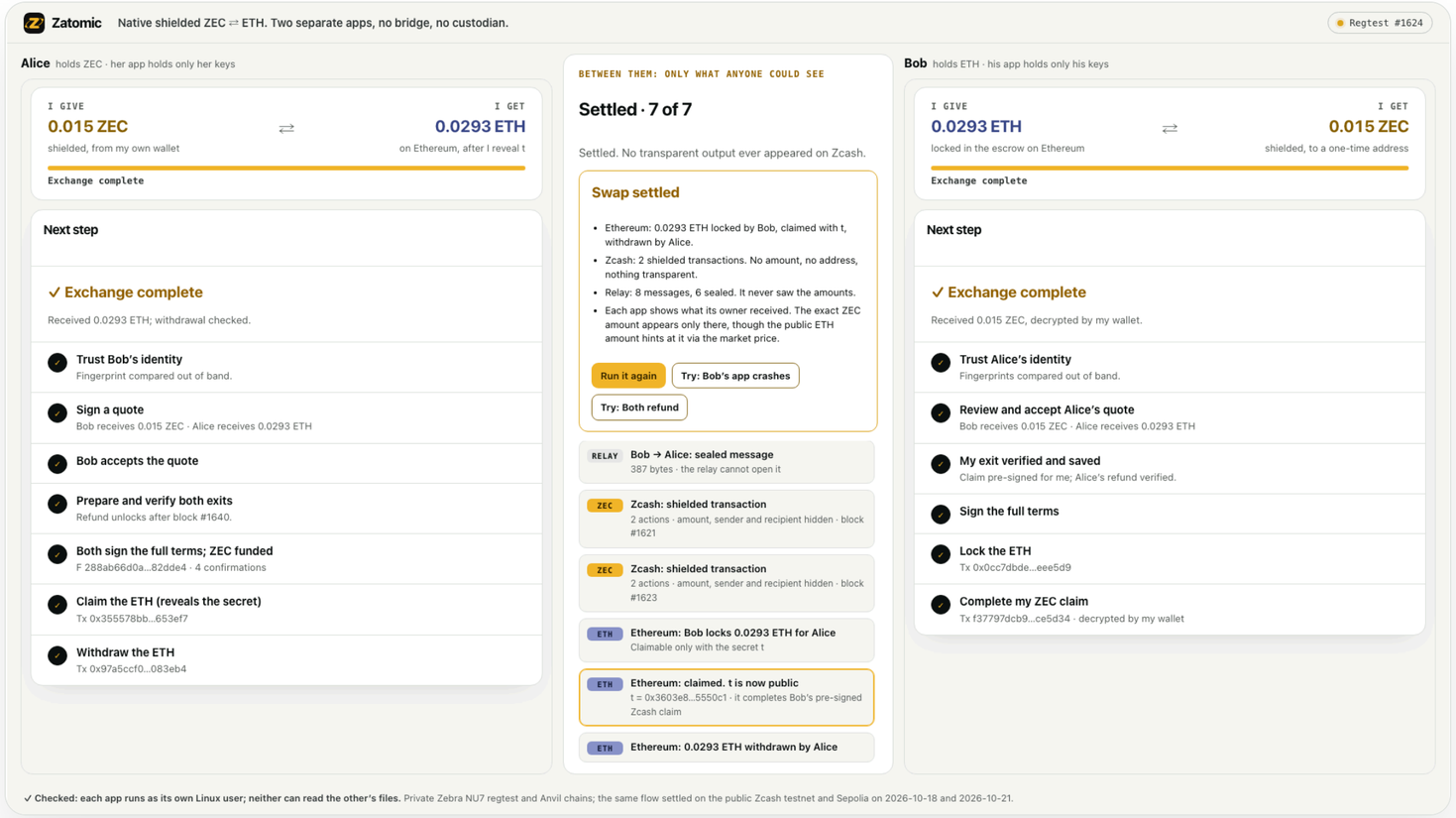
Signed before the note exists: in Zcash v6, signatures do not cover the anchor or the proof.

A settled swap stays fully shielded. Only a refund uses one small transparent coin from Alice’s own change, because Zcash enforces a lock time only there.

LIVE DEMO

Two separate apps. Only public data between them.

Real Zebra NU7 and Anvil transactions on private test chains. Each app holds only its owner’s keys.



PUBLIC TESTNETS · ZCASH TESTNET + SEPOLIA

Both outcomes, on real networks

4

funded public runs: two settled, two refunded both sides

2

Zcash transactions in a settled swap, both fully shielded

−79%

Ethereum gas per settled swap with the shipped V2 escrow

Every transaction is linked in the records: github.com/yugocabrio/zatomic

WHAT IS NEW

Zcash’s own spend authorization, as one side of an atomic swap

- A RedPallas adaptor signature on a 2-of-2 FROST key
- Ethereum checks t on Pallas directly: no bridge, no cross-curve proof
- If the trade fails, Zcash itself returns Alice’s ZEC

`Live demo · 162-55-186-88.sslip.io` `Code · github.com/yugocabrio/zatomic`